

Discrete Vortical Gust Encounter and Mitigation using Closed Loop Control

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The open and closed-loop response of a wing to discrete vortical gusts of differing strengths was characterized using Time Resolved Particle Image Velocimetry (TR-PIV) in the University of Dayton Water Tunnel (UD-WaT). Discrete vortical gusts were generated by rapidly pitching a flat plate gust generator upstream of the wing. Simultaneous Time Resolved Particle Image Velocimetry (TR-PIV) and force measurements on the flat plate wing downstream of the gust generator were collected. A closed loop Proportional Integral Derivative (PID) control system was implemented in order to mitigate the influence of the discrete gust on the wing. TR-PIV was used to correlate the flow physics responsible for the force histories experienced by the wing. Peak-to-peak and average gust mitigation using closed loop control were quantified based on the deviations from open-loop results. The controller reduced the force transients during the gust entry by an average of 67% whereas the gust exit transients were only reduced by an average of 6%.

Nomenclature

A	=	Oscillation amplitude (degrees)	S	=	Shear (1/s)
u	=	x component of velocity (m/s)	C_L	=	Coefficient of Lift
U_∞	=	Free stream velocity (m/s)	Abbreviations		
v	=	y component of velocity (m/s)	FOV	=	Field of View
t^*	=	Convective time	TR-PIV	=	Time Resolved Particle Image
$\dot{\alpha}$	=	Pitch Rate (degrees/s)	Velocimetry		
α_N	=	Normalized Pitch Rate	UD-WaT	=	University of Dayton Water Tunnel
Q	=	Q-Criterion (1/s ²)			
ω	=	Vorticity (rad/s)			

I. Introduction

The study of gusts has become an increasingly important field in aerodynamics due to the growth and widespread application of small-scale, low-speed air vehicles. The use of unmanned aerial vehicles (UAVs) has expanded across a diverse range of sectors, including defense, agriculture, recreation, and beyond. These small-scale UAVs fly under unsteady environments where aerodynamic disturbances greatly impact the stability and performance of an aerial vehicle. [1] These aerodynamic disturbances are categorized as transverse, streamwise, or vortical gusts.[2] Vortical gusts can be generated with periodic or discrete gust profiles.[3] This study will focus on the generation, characterization, and control of discrete vortical gusts.

Vadher and Babinsky [4] showed that vortical gusts of varying strengths can be created by rapidly pitching a symmetric airfoil. In the study, a symmetric airfoil was towed through a tank and pitched rapidly to create a vortical gust. The present study will use a similar method of gust generation but will, instead, involve rapid pitching of a flat plate. The purpose in using a flat plate rather than an airfoil is to reduce the effect of the shear layer wake impingement on the downstream wing. The gust will travel downstream where its interaction with a wing will be investigated using TR-PIV and force measurements.

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Numerous studies suggest that implementing a closed-loop control system is more effective than open-loop mitigation of streamwise, transverse, or vortical gusts. Sedky et al. [5] were able to mitigate transverse gusts with an open-loop control system based on a Goman Khrabrov model. This model predicted the effective angle of attack and used pitching maneuvers during the gust encounter to mitigate C_L transients. When employed in a closed-loop control simulation, the model effectively mitigated the C_L transients. However, the open-loop control experiments failed to effectively mitigate the changing lift following the gust encounter. The alternating leading and trailing edge vortices require the use of a closed-loop control system to effectively track the C_L transients throughout the encounter, namely in the gust exit transient.

Herrmann et al. [6] developed a Proportional-Integral (PI) control system actuating a trailing edge flap to control a periodic gust profile. The PI system was paired with a model-based feed forward system to better mitigate the effects of the gust profile. Mongin et al. [7] developed a PID-based closed-loop controller tailored for unsteady gusts at high amplitudes. Killian et al. [8], [9] used the same PID-based controller as Mongin et al. [7] to investigate the effectiveness of a closed-loop controller on sinusoidal periodic gust profiles. They found that effective mitigation can be achieved during the entrance transient of the gust encounter. However, as the gust leaves the wing, the wing experiences an unmanageable C_L fluctuation. Hence, this study will investigate the effectiveness of a closed-loop controller on a discrete gust to isolate the entry and the exit transients.

In this study, the characterization for a series of gusts of different strengths and their impact on a wing placed downstream of the gust generator will be shown. A flat plate gust generator will be rapidly pitched at three different amplitudes and two different pitch rates. TR-PIV will be used to track the vorticity and vortex centers as each gust moves through the FOV. The gusts will impinge on the wing placed in the FOV, showing flow interaction with the wing in open-loop. A PID controller will be applied to hold a constant C_L during the gust encounter. The force histories of the open- and closed-loop tests will be compared to determine the effectiveness of the PID controller in mitigating the impact of the gusts on the wing. TR-PIV will be used to show the differences in flow physics between the open- and closed-loop cases.

II. Experimental Setup

A. University of Dayton Water Tunnel

All experiments were performed at the University of Dayton Water Tunnel, shown in Fig. 1a. For a detailed explanation of the experimental setup as well as calibration of the tunnel speed and turbulence intensity refer to Mongin et al. [7].

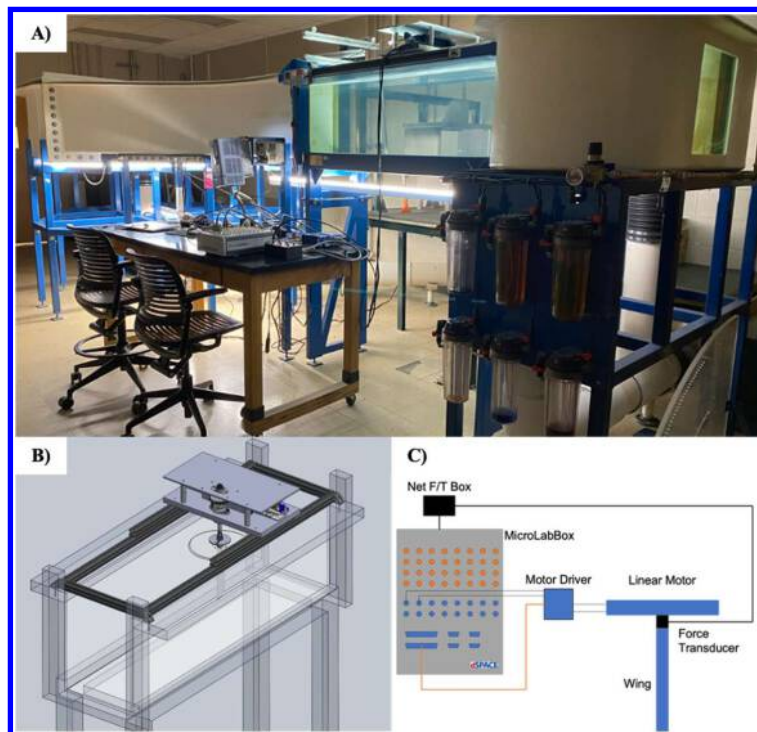


Figure 1 A) University of Dayton Water Tunnel B) CAD model of wing mount and linear motor C) Wiring diagram of MicroLabBox and wing control system

The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm length. The free-surface speed range is 7.5 cm/s to 38 cm/s. For this study, the freestream velocity of the tunnel was set to 0.28m/s and a flat plate with a 3-inch (0.076m) chord and 6-inch (0.152m) span was used. The tunnel is equipped with a one degree of freedom motion stage consisting of a H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. This motor is outfitted with a rack and pinion gear that converts its linear motion to a rotational motion in order to pitch a test article in the tunnel test section as shown in Fig. 1b.

The closed-loop feedback and control for this system is performed with a dSPACE MicroLabBox with BNC type terminations using the MATLAB LabVIEW Real Time Interface (RTI) programming environment. The MicroLabBox is capable of up to 24 channels of analog input and 16 channels of output in 16-bit at 1 Msps. There are also 48 channels of bidirectional digital inputs and outputs available. The MicroLabBox was ran at 1kHz for these experiments although it is capable of up to 2Ghz processing in real-time. Force measurements were performed with an ATI Mini40 force transducer with NET F/T box interfaced with the dSPACE equipment using the ethernet protocol blocks in Simulink. A wiring schematic of the feedback system is shown in Fig. 1c. The force transducer F/T outputs are taken into the MicroLabBox and used to calculate aerodynamic forces on the test article. The motor driver is connected to two analog outputs, one for enabling the motor and the other for commanding the current of the motor. Another connection between the motor driver and MicroLabBox is made via the digital I/O ports to receive information from the motor encoder.

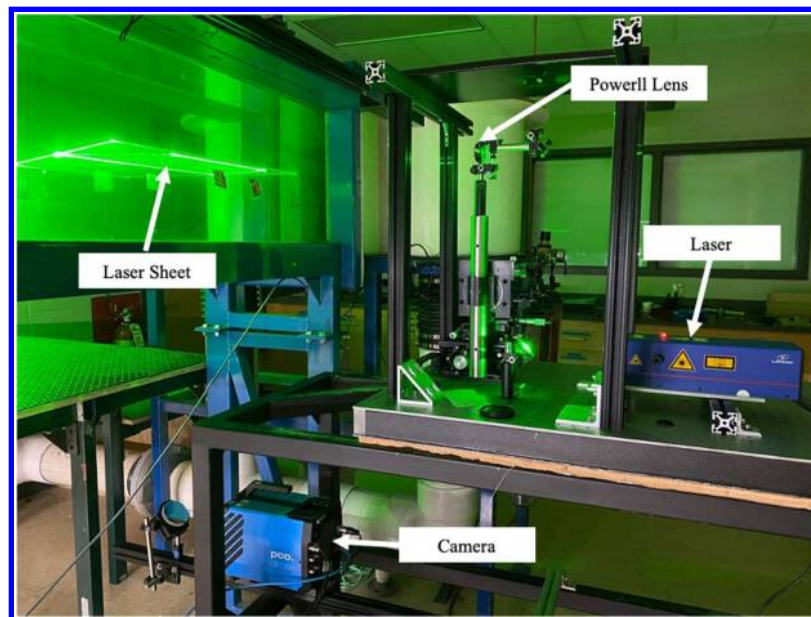


Figure 2 UD-WAT PIV system

C. Particle Image Velocimetry

The gust characterization and changes in the flowfield around the wing in both open and closed loop cases were captured through time resolved PIV. The PIV system shown in Fig. 2 is comprised of a PCO DiMax S4 high speed camera with a 2016x2016 (4 Megapixel) CMOS sensor and a 3-Watt continuous wave laser from LaVision.

Table 1 PIV Test Matrix

Type	A ($^{\circ}$)	α_N ($^{\circ}$)
Gust Characterization	20, 30	30, 45
Open Loop	20, 30	30, 45
Closed Loop	20, 30	30, 45

The field of view is oriented from the bottom of the test section facing up, requiring the camera to pass through a 45-degree mirror. The laser beam is passed through a series of mirrors and into a Powell lens and projected as a light

sheet with a 30 degree fan angle. 60 μ m polyamide seeding particles from LaVision are used in the tunnel to be illuminated by the laser. Triggering of the camera recording is done with the MicroLabBox and the trigger signal is recorded along with the time locked data from the sensor and the pitch position of the test article. This ensures that the PIV data and the MicroLabBox data are time synced. Post processing of the PIV image frames was done with the Fluere software from Kyle Lynch [10]. The post processed PIV data and the data collected from the MicroLabBox are combined and analyzed using MATLAB. The test matrix for the gust characterization is shown in Table 1.

C. Vortical Gust Generator

The vortical gust generator was previously used in a study by Killian et al [8], [9] and is a flat plate with a span of 18.5 in. and a 3 in. chord. The flat plate has a thickness of 0.125 in. The gust generator is mounted to the stepper motor and pivots on a shaft held in place by a magnetic post. The gust generator spans from the base of the tunnel to the free surface of the test section. As shown in Fig. 3b, the quarter chord of the gust generator was mounted 5 chords upstream of the quarter chord mounting position of the flat plate wing. A schematic of the gust generator pitched at a desired amplitude can be seen in Fig. 3a. The gust generator was rapidly pitched according to a pair of normalized pitch rates determined from Equation 1 where $\dot{\alpha}$ is the pitch rate, c is the chord, and U_∞ is the freestream velocity. Dimensional pitch rates of 110 $^\circ$ /s and 165 $^\circ$ /s resulted in normalized pitch rates of 30 and 45.

$$\alpha_N = \frac{\dot{\alpha}c}{U_\infty} \quad (1)$$

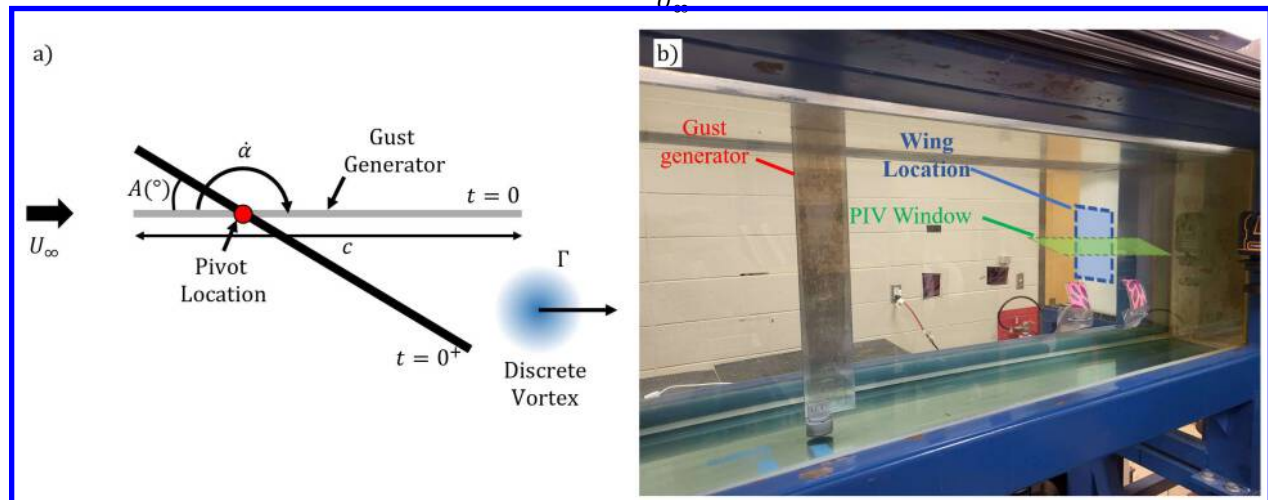


Figure 3 Gust Generator and PIV Window

D. Control System

The closed loop control system was previously used in a study by Mongin et al [7] and the schematic of the controller is shown in Fig 4. For the current study, a C_L of 0 was set as a target for each of the closed-loop cases. The gust was allowed to convect downstream while real-time-sensed C_L values from an ATI Mini40 force sensor were used to determine the error between the commanded C_L and the sensed C_L . The calculated error was sent to a PID controller, which determines a commanded AOA sent to the linear motor controlling the flat plate wing. The wing, then, experiences a new C_L , which feeds back into the closed loop.

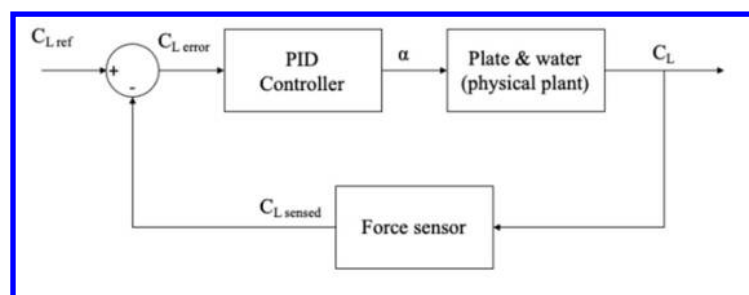


Figure 4 Closed loop control system

III. Results

A. Gust Characterization

Discrete vortical gusts were generated by rapidly pitching a flat plate to a series of set amplitudes at predetermined pitch rates. Fig. 5 shows the vorticity of the three gusts from each pitch amplitude at a dimensional pitch rate of $165^\circ/\text{s}$. Each plot spatially unwraps the vorticity contours from each frame acquired. The right side of the plot shows the undisturbed flow prior to any deflection of the gust generator. This region lacks any large disturbances to the freestream velocity. Each gust is shown in the frame along with a dashed line representing the wing location. The wing was not present for the gust characterization, but it is important to note the distance between the vortex and the wing location. A vortex offset from the wing in either direction will result in a different behavior when compared to a symmetric impingement. The region to the left of the gust shows a line of sporadic vorticity before larger shedding from the pitched gust generator enters the field of view (FOV). This line of vorticity is weak in magnitude and significantly offset from the wing location. This should have a minimal effect on C_L prior to the large region of shedding on the left side of Fig. 5.

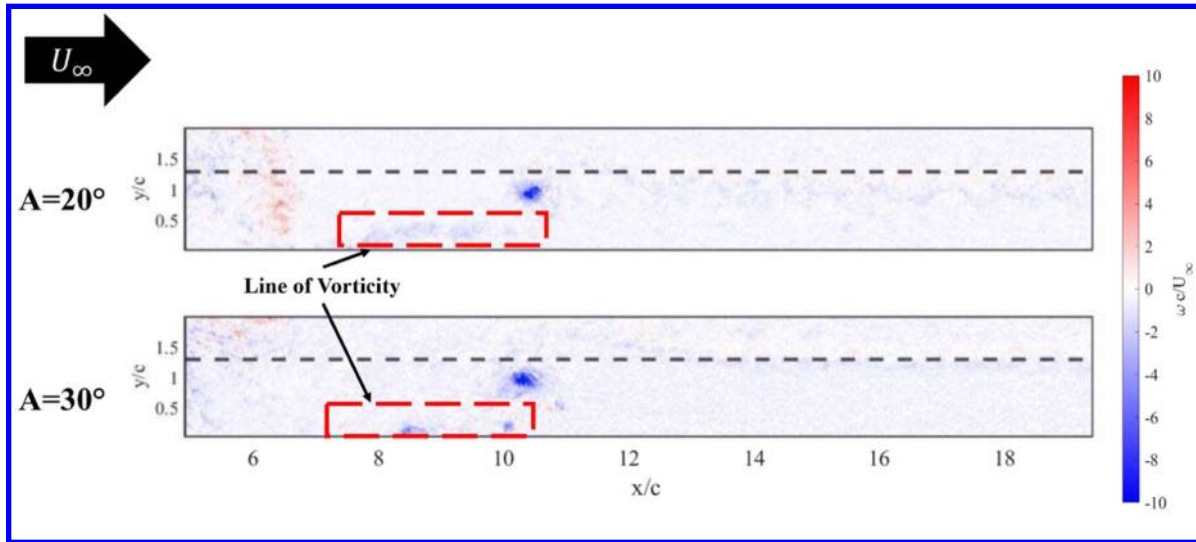


Figure 5 Gust canvas of discrete vortices generated at a dimensional pitch rate of $165^\circ/\text{s}$. The dashed lines represent the location where the wing will be placed.

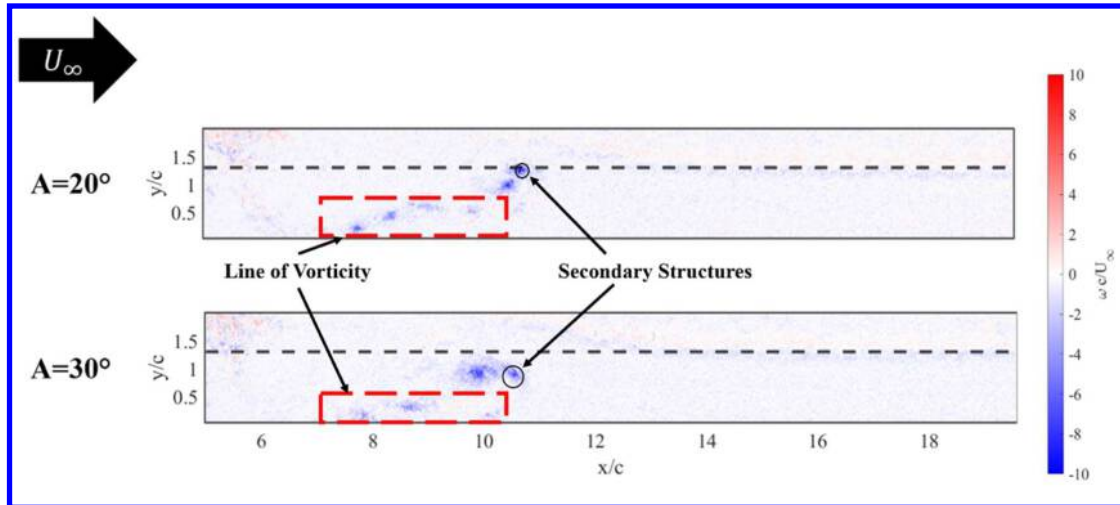


Figure 6 Gust canvas of discrete vortices generated at $110^\circ/\text{s}$

Fig. 6 shows the same spatially unwrapped gust profiles at a dimensional pitch rate of $110^\circ/\text{s}$. The slower pitch rate shows less coherent gusts, as well as the presence of more prominent secondary structures in the flow. The slower dimensional pitch rate cases produce weaker gusts. The slower pitch rate does not produce a region of vorticity

on the same magnitude as the cases with a dimensional pitch rate of $165^\circ/\text{s}$. The radii of the vortices in the slower pitch rate cases are smaller than the vortex radii in Fig.5. Fig. 6 shows the weaker vorticity and the rapid decay of vorticity moving away from the vortex center

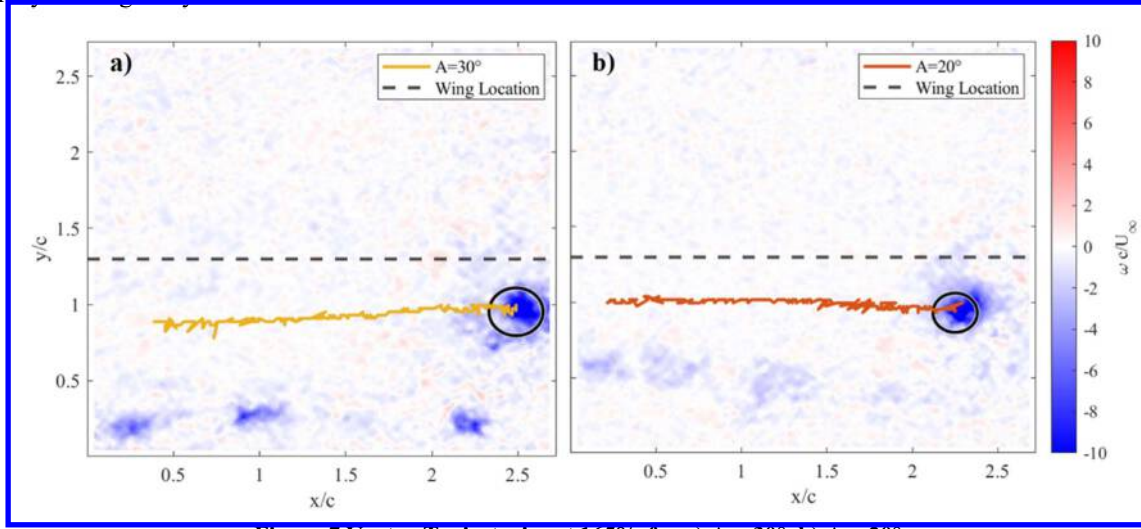


Figure 7 Vortex Trajectories at $165^\circ/\text{s}$ for a) $A = 30^\circ$, b) $A = 20^\circ$

The primary aim of the gust characterization is to accurately track the path and intensity of the vortex as it progresses across the FOV. The vortex core was identified using the maximum Q-criterion shown in Equation (2). Here, ω denotes vorticity, and S represents shear. These parameters were computed through finite difference methods applied to the partial derivatives, as outlined in Equations (3) and (4). In the majority of the Particle Image Velocimetry (PIV) frames, the discrete vortices were clearly identifiable via the Q-criterion.

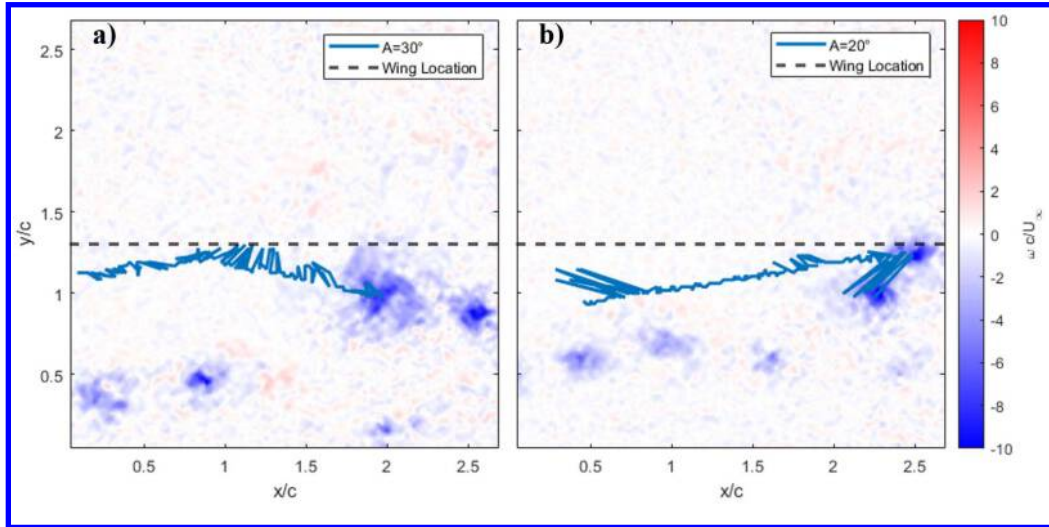


Figure 8 Vortex Trajectories at $110^\circ/\text{s}$ for a) $A = 30^\circ$, b) $A = 20^\circ$

The maximum Q-criterion was consistently logged as the vortex center at each time step. In certain instances, outliers in the x- and y-coordinates were identified and rectified using MATLAB's filloutliers() function, employing linear interpolation between adjacent non-outlier points. The y-coordinate outliers were identified using a Grubbs test, while the x-coordinate outliers were detected through a moving median approach with a sample window of 10.

$$Q = \frac{1}{2} (\|\omega\|^2 - \|S\|^2) \quad (2)$$

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad (3)$$

$$S = \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \quad (4)$$

Fig. 7 shows the vortex tracking for the 165°/s case. Each of the vortex trajectories are offset from the wing location. This gap trends upward with increasing magnitude of deflection. The trajectory of the 30° case drifts back toward the centerline of the FOV, where the wing is placed. The 20° case does not track back toward the wing and drifts away from the centerline of the test section. Multiple smaller regions of vorticity with less strength are shown in each case. These small regions of vorticity are shed off the gust generator during and after the pitching motion, but each lacks the coherence and strength of the primary vortex. Fig. 8 shows the vortex tracking for the 110°/s case. The trajectory for the slower dimensional pitch rate case is harder to track due to the secondary vortices in the flow. As the vortex moves through the FOV, the calculated vortex center is shifted due to a nearby secondary structure.

B. Gust Mitigation

A flat plate wing with an effective aspect ratio of 4 was placed in the FOV where the gust was characterized, and open- and closed-loop tests were conducted to quantify the impact of the wing gust encounter. The gust mitigation was quantified with both peak-to-peak percent mitigation, as well as average mitigation across the gust encounter. The peak-to-peak mitigation was calculated according to Equation (4), with a target C_L of 0. The peak values are the maximum and minimum values from the positive and negative side of the gust encounter for the open- and closed-loop cases. The average error in Equation (5) is the average difference between the target C_L and the sensed C_L throughout the case. The average mitigation in Equation (6) compares the average C_L error in open and closed loop calculated in Equation (5). The “Gust Entry Mitigation” in Table 2 refers to the positive C_L peak experienced by the wing as the gust approaches the leading edge. The “Gust Exit Mitigation” refers to the negative C_L peak experienced by the wing as the gust crosses the trailing edge of the wing.

$$mit_{peak2peak} = 1 - \frac{(C_{L_{peak\ closed}} - C_{L_{target}})}{(C_{L_{peak\ open}} - C_{L_{target}})} \quad (4)$$

$$E_{avg} = \frac{1}{n} \sum |C_{L_{target}} - C_{L_{sensed}}| \quad (5)$$

$$mit_{avg} = 1 - \frac{E_{avg\ closed}}{E_{avg\ open}} \quad (6)$$

The gust entry mitigation is consistently higher than the gust exit mitigation in all cases. The PID controller is able to consistently achieve greater than 60% mitigation during gust-entry. However, as the gust travels across the wing and exits, there is increased difficulty in the mitigation of the lift transients. Despite the drastic differences in the peak-to-peak mitigation, the average mitigation across the entire encounter consistently reaches greater than 45%. The flow physics at each peak of the open- and closed-loop tests, along with the C_L history, is shown in Fig. 11 for the dimensional pitch rate of 165°/s with an amplitude of 30°.

Table 2 Gust Mitigation

Amplitude	Pitch Rate	Gust Entry Mitigation	Gust Exit Mitigation	Average Mitigation
30°	165°/s	67.55%	1.84%	45.71%
	110°/s	69.27%	5.09%	48.06%
20°	165°/s	68.73%	8.12%	48.96%
	110°/s	67.75%	7.87%	48.04%

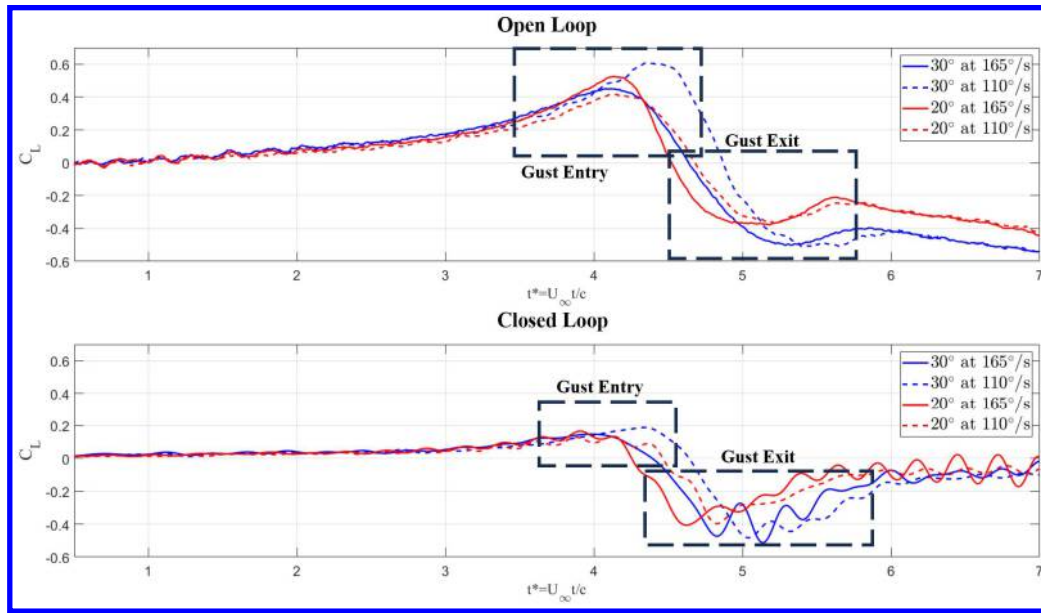


Figure 9 Gust mitigation comparison between open and closed loop cases at different α_N

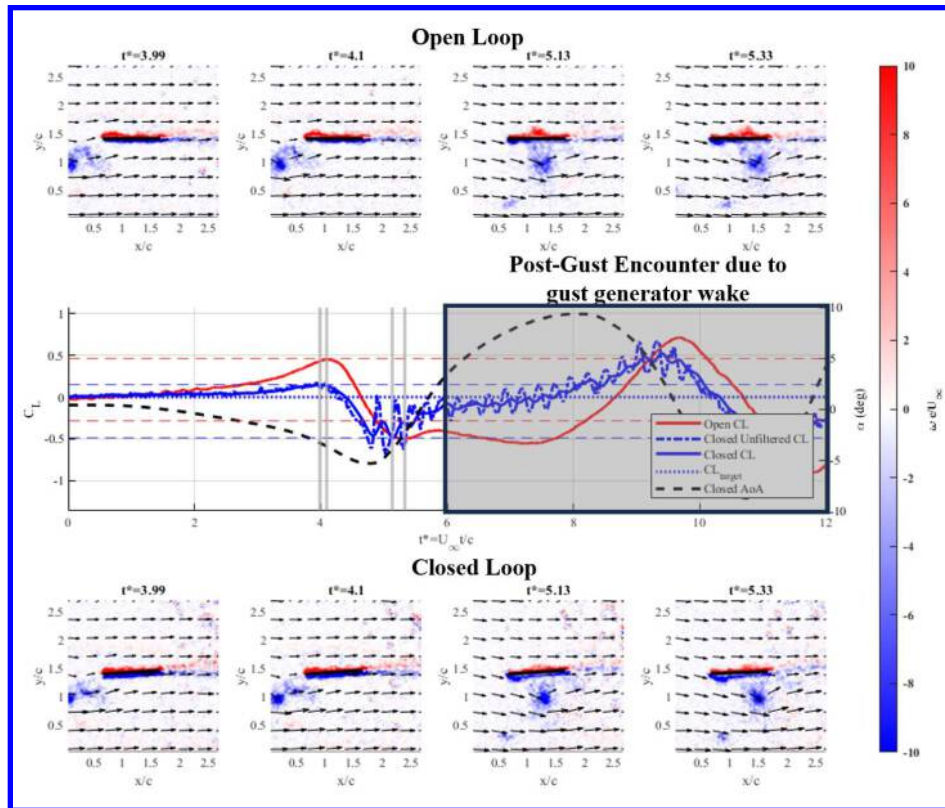


Figure 10 Full Test Gust Mitigation for $A = 30^\circ$ and $\alpha = 165^\circ/\text{s}$

The open and closed loop C_L response can be seen in Fig 9. The 30° case at a dimensional pitch rate of $110^\circ/\text{s}$ has the highest peak C_L of any case. Even though the gust in this case is weaker than the $165^\circ/\text{s}$ case, the trajectory of this gust follows closer to the wing, increasing the effect of the gust. This case experiences a similar peak during the gust exit as well. The $165^\circ/\text{s}$ cases follow a similar trend in the gust entry for both the open and closed loop cases. However, during the gust exit, Fig 9. shows a later and larger exit peak for the 30° case. Any differences between cases during the gust entrance are exaggerated during the gust exit. The closed loop peaks in each case are consistently earlier in the gust encounter when compared to the open loop equivalents.

The full data for the 30° amplitude at $165^\circ/\text{s}$ pitch rate is shown in Fig. 10. The averaged, unfiltered C_L for the case is shown. The high frequency oscillations in C_L make it difficult to calculate peak to peak mitigation. A Butterworth filter was applied to each case to reduce the noise in the closed loop case. The mitigation for each case was calculated from the filtered C_L . The initial effect of the gust is seen at each frame shown. The wake of the gust generator causes additional changes in C_L that were not desired. The shaded region of Fig 10. is excluded from further analysis to focus on the gust encounter while ignoring gust generator wake effects. To counter this, future studies will be conducted with a ramp and return profile instead of the pitch and hold profile used in this study.

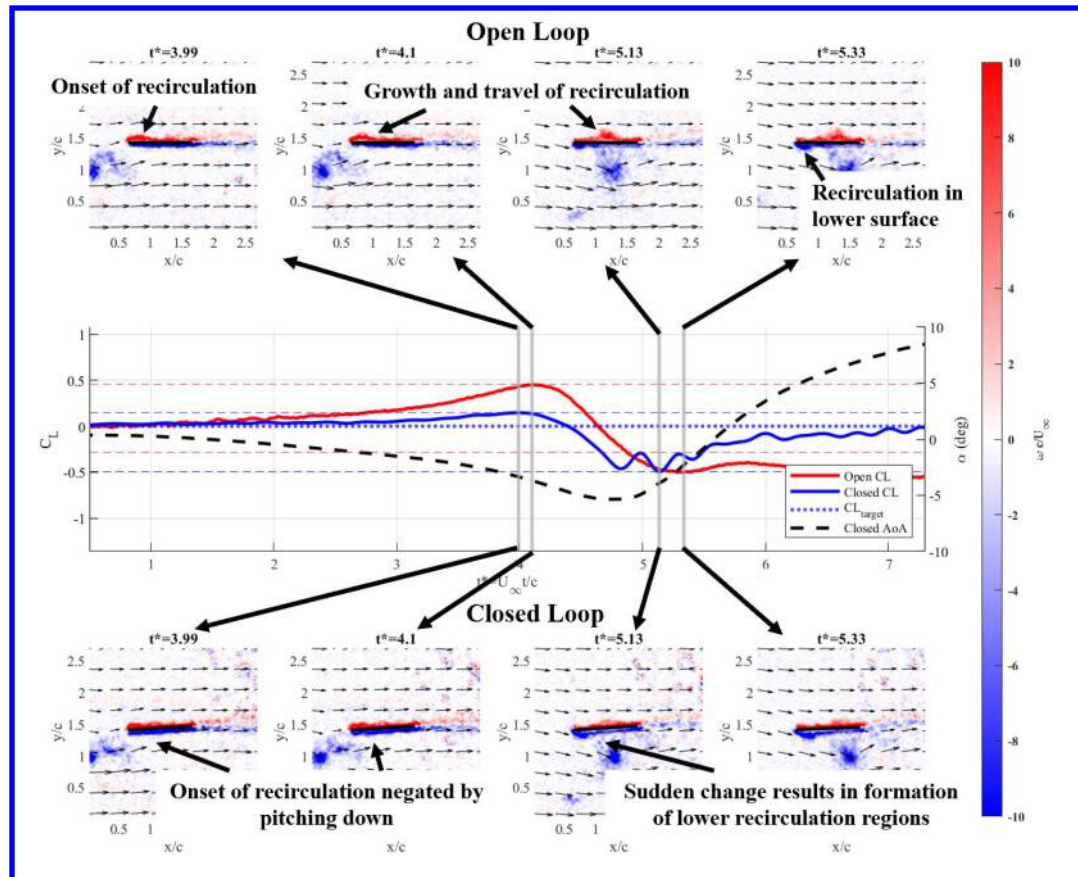
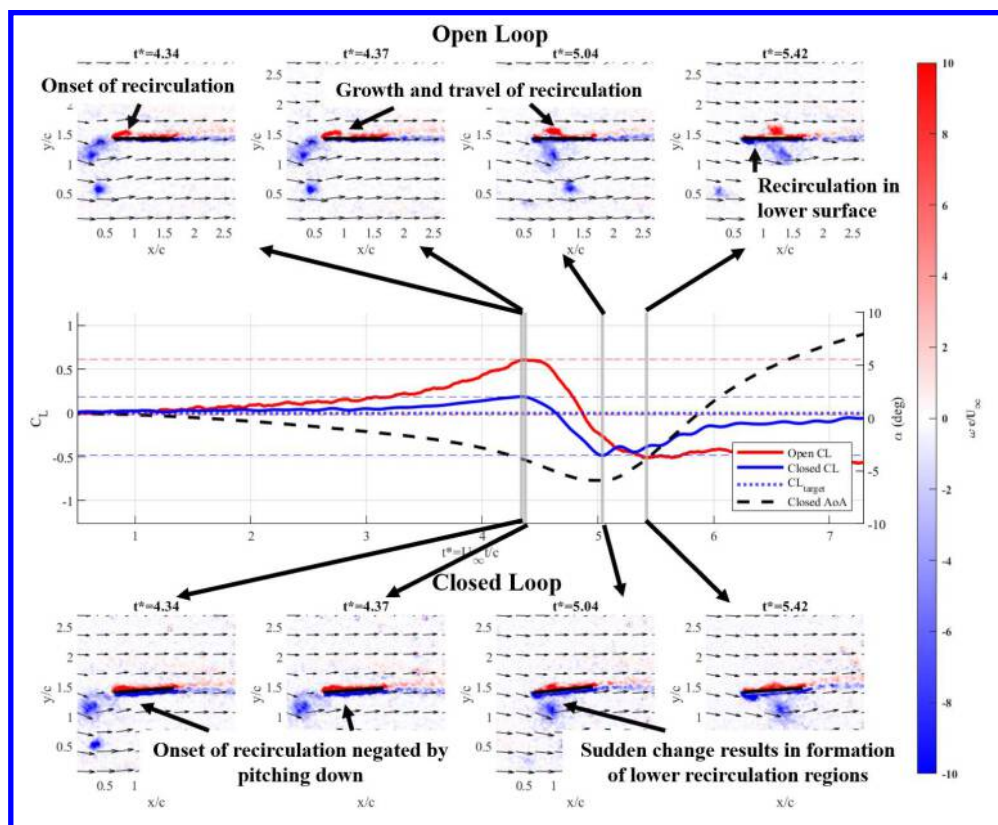
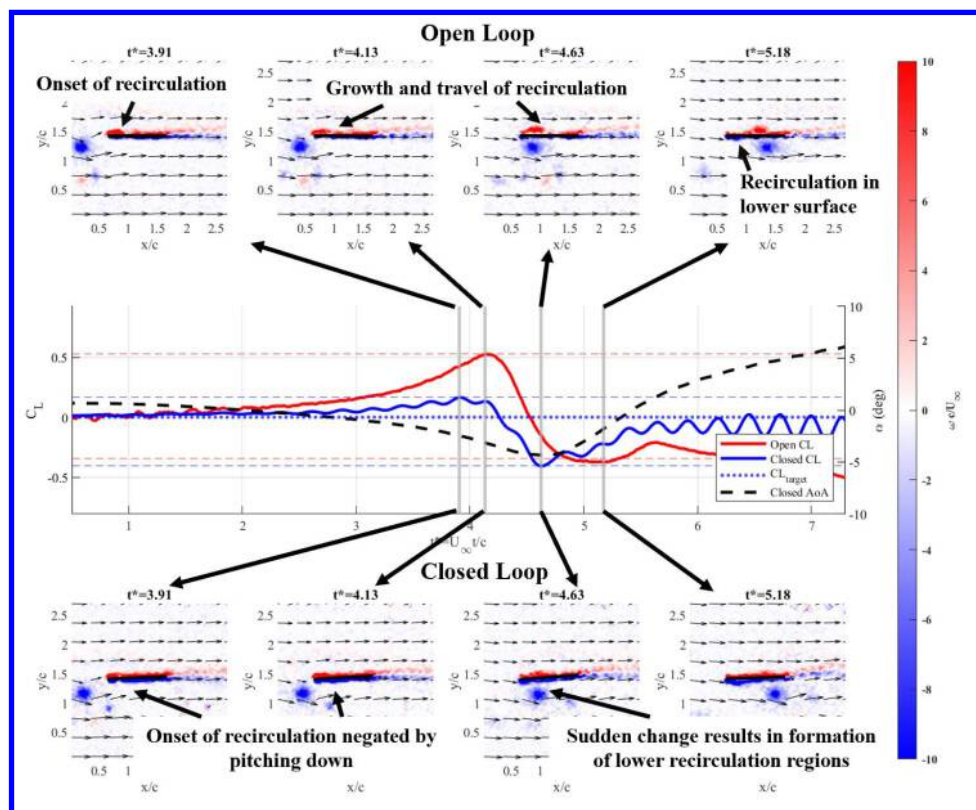


Figure 11 Gust Mitigation for $A = 30^\circ$ and $\alpha = 165^\circ/\text{s}$

The first frame shown at $t^* = 3.99$ shows the flow at the beginning of the gust encounter. At this convective time, even though the flow field in both open and closed-loop cases look identical, a slight onset of a recirculation region can be observed in the open-loop case resulting in an increased C_L transient. The second frame shown at $t^* = 4.1$ corresponds to the maximum C_L peaks observed in open-loop case. As the vortex approaches the wing, the recirculation region grows further, increasing the C_L . However, in the closed loop case, the wing pitched down to mitigate the transients experienced. As such, no visible recirculation region was formed. The third frame shows the gust around the mid chord of the wing in each frame. The recirculation region on the open and closed-loop cases has shed off the leading edge and can be seen tracking with the discrete vortex. The shedding of the recirculation region as the gust travels across the lower surface of the wing causes the sharp drop in C_L from the positive peak to the negative peak. The final frame shows the negative peak of the open-loop case. The gust has moved out of the FOV, and a recirculation region has formed on the lower surface of the wing. This induces the opposite effect experienced by the wing in the first frame.

Figure 12 Gust Mitigation for $A = 30^\circ$ and $= 110^\circ/s$ Figure 93 Gust Mitigation for $A = 20$ and $= 165^\circ/s$

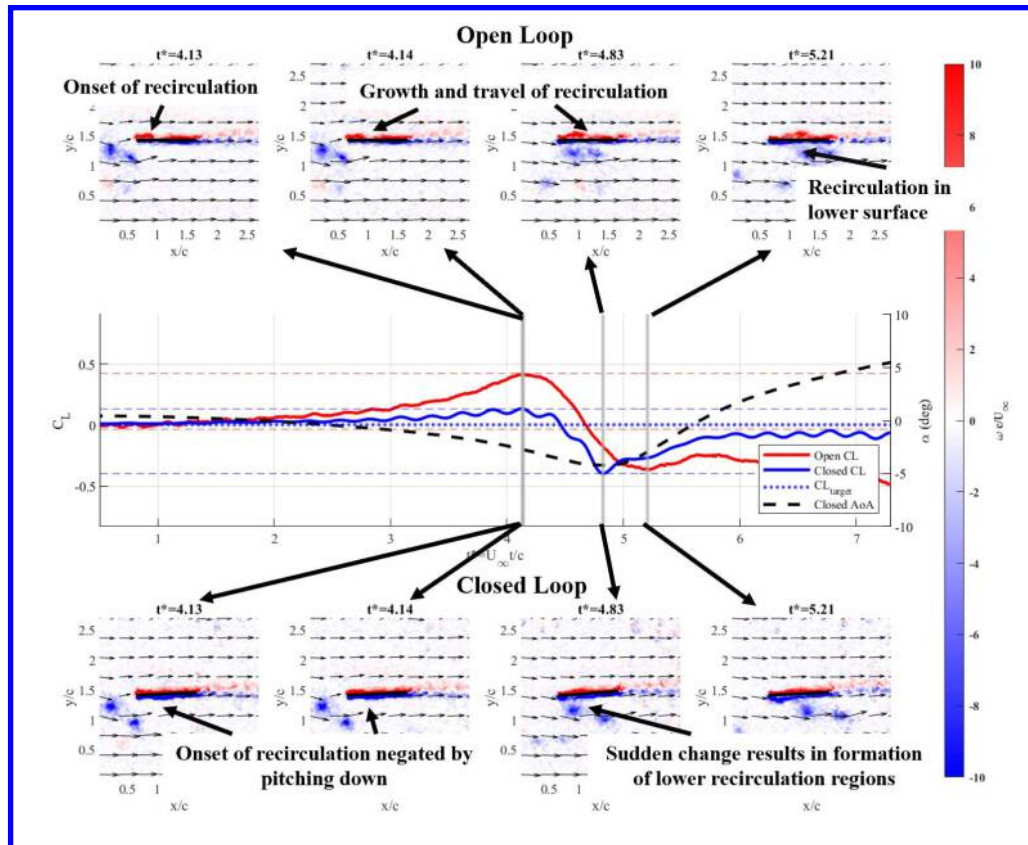


Figure 14 Gust Mitigation for $A = 20^\circ$ and $\alpha = 110^\circ/\text{s}$

The same frames are shown in Fig. 12 for the 30° at $110^\circ/\text{s}$ cases. The first frame shown at $t^* = 4.34$ shows the flow at the beginning of the encounter. The onset of a recirculation region on the upper surface of the wing can be seen again. The second frame at $t^* = 4.37$ corresponds to the maximum open loop C_L experienced by the wing. The closed loop frame at this time does not show a large recirculation region like in the open loop case. The wing pitched down, mitigating the transients experienced in the open loop case. The third frame shows the gust around the mid chord of the wing in each frame. The recirculation region on the open- and closed-loop cases has shed off the leading edge and can be seen tracking with the discrete vortex. The same drop in C_L seen in the $165^\circ/\text{s}$ case can be seen as the recirculation region travels down the chord of the wing. A recirculation region can then be seen at $t^* = 5.42$ in the open and closed loop cases causing the C_L error experienced in both open and closed loop cases.

The same frames are shown in Fig. 13 for the 20° at $165^\circ/\text{s}$ cases. The first frame shown at $t^* = 3.91$ shows the flow at the beginning of the encounter. A recirculation region on the upper surface of the wing can be seen in the open loop frame. The second open loop frame at $t^* = 4.13$ shows a larger recirculation region as the gust approaches the wing. The closed loop frame shows a pitched wing reducing the recirculation region mitigating the C_L peak. The third frame shows a recirculation region forming on the lower surface of the wing. The sudden change from an upper surface recirculation region to a lower surface recirculation region during the gust exit makes it difficult to control.

The same frames are shown in Fig. 14 for the 20° at $110^\circ/\text{s}$ cases. The first frame shown at $t^* = 4.13$ shows the flow at the beginning of the encounter. The onset of a recirculation region on the upper surface of the wing can be seen. The second frame at $t^* = 4.14$ corresponds to the maximum open loop C_L experienced by the wing. The closed loop frame at this time does not show a large recirculation region like in the open loop case. The wing pitched down, mitigating the transients experienced in the open loop case. The third frame shows the gust around the mid chord of the wing in each frame. The recirculation region on the open and closed loop cases has shed off the leading edge and can be seen tracking with the discrete vortex. This frame also shows a significant decay in vorticity for the secondary vortex on the lower surface of the wing. The closed loop frame shows a stronger gust remaining than the open loop case at $t^* = 4.83$. The same drop in C_L seen in the $165^\circ/\text{s}$ case can be seen as the recirculation region travels down the chord of the wing. A recirculation region can then be seen at $t^* = 5.21$ in the open and closed loop cases causing the C_L error experienced in both open and closed loop cases.

IV. Conclusions

Discrete vortical gusts were generated, and their impact on a downstream wing were investigated using synchronized TR-PIV and force measurements. The force histories and flow physics were compared to determine the impact of the gust on the wing. The PID controller used in the study was able to achieve a maximum average C_L mitigation of 48.83% and an average gust entry C_L mitigation of 67%. However, the controller performed significantly worse during the gust exit transient. The maximum gust exit mitigation was 6%. A recirculation region was formed on the upper surface of the wing, which resulted in a large C_L transient during the gust entry in the open-loop case. Once the vortex crossed the half-chord mark, the wing experienced a large flip from positive to negative C_L as expected. The extent of recirculation growth and shedding was reduced by the closed-loop controller, in an effort to maintain a target C_L of 0. To reduce the effect of the shear layer wake of the gust generator, in future work, a ramp return gust profile will be implemented, while maintaining the isolated gust in the flow. In order to achieve better gust exit mitigation, a more aggressive PID tuning will also be pursued. It is important to balance the aggressiveness of the tuning with the stability necessary to track the desired C_L in weaker gusts. A newly trained PID controller will need to track the gradual C_L changes leading up to the encounter, while also being able to closely track the sharp change in C_L experienced after the recirculation region has been shed. Other methods of mitigating this part of the encounter would require different control devices or control surfaces on the wing. Trailing edge devices, morphing wing structures, or feed forward control systems could all contribute to mitigating the exit transient of the gust.

Acknowledgments

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